



qualcomm/**Phi-4-Mini-Instruct**



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⚡ Inference Providers

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Text Generation

This model isn't deployed by any Inference Provider.



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Paper for qualcomm/Phi-4-Mini-Instruct

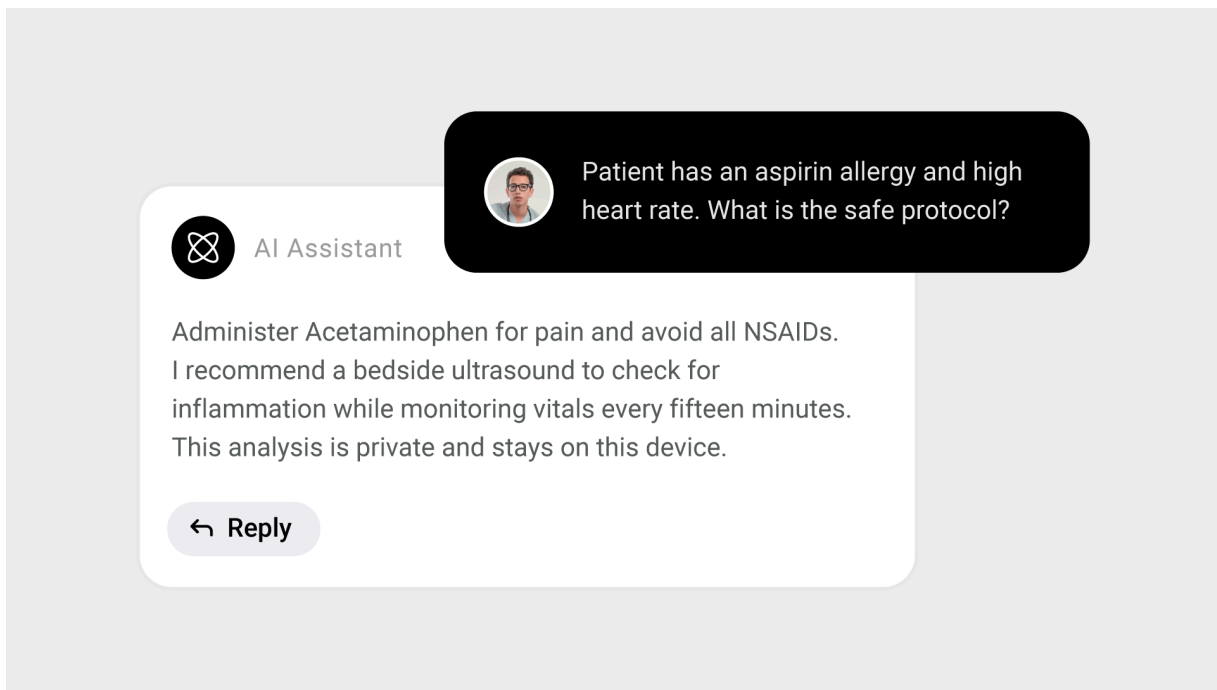
Phi-4 Technical Report



Paper • 2412.08905 • Published Dec 12, 2024 • 124



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🔗 **Phi-4-Mini-Instruct: Optimized for Qualcomm Devices**

Phi-4-mini-instruct is a lightweight open model built upon synthetic data and filtered publicly available websites - with a focus on high-quality, reasoning dense data.

This is based on the implementation of Phi-4-Mini-Instruct found [here](#). This repository contains pre-exported model files optimized for Qualcomm® devices. You can use the [Qualcomm® AI Hub Models](#) library to export with custom configurations. More details on model performance across various devices, can be found [here](#).

Qualcomm AI Hub Models uses [Qualcomm AI Hub Workbench](#) to compile, profile, and evaluate this model. [Sign up](#) to run these models on a hosted Qualcomm® device.

🔗 **Deploying Phi-4-Mini-Instruct on-device**

Follow the [GenieX quickstart](#) to install GenieX and deploy the model on a target device.

🔗 **Getting Started**

There are two ways to deploy this model on your device:

Option 1: Download Pre-Exported Models

Below are pre-exported model assets ready for deployment.

Runtime	Precision	Chipset	SDK Versions	Download
GENIEX_LLAMACPP	q4_0	Universal		Download

For more device-specific assets and performance metrics, visit [Phi-4-Mini-Instruct on Qualcomm® AI Hub](#).

Option 2: Export with Custom Configurations

Use the [Qualcomm® AI Hub Models](#) Python library to compile and export the model with your own:

- Custom weights (e.g., fine-tuned checkpoints)
- Custom input shapes
- Target device and runtime configurations

This option is ideal if you need to customize the model beyond the default configuration provided here.

See our repository for [Phi-4-Mini-Instruct on GitHub](#) for usage instructions.

Model Details

Model Type: Model_use_case.text_generation

Model Stats:

- Number of parameters: 3.8B

- **TTFT: Time To First Token** is the time it takes to generate the first response token. This is expressed as a range because it varies based on the length of the prompt. The lower bound is for a short prompt (up to 128 tokens, i.e., one iteration of the prompt processor) and the upper bound is for a prompt using the full context length (4096 tokens).
- **Response Rate:** Rate of response generation after the first response token.

[🔗 Performance Summary](#)

Model	Runtime	Precision	Chipset	Context Length	Response Rate (tokens per second)	Time To First Token (range, seconds)
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® 8 Elite Gen 5 Mobile	512	23.272858	0.92645325 - 1.03434575
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® 8 Elite Gen 5 Mobile	512	22.664968	1.03434575 - 1.17659049
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® 8 Elite Gen 5 Mobile	512	19.242405	0.17659049 - 0.70636199
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® 8 Elite Gen 5 Mobile	4096	12.621046	2.25159937 - 72.0511799
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® 8 Elite Gen 5 Mobile	4096	12.03971	2.47942718 - 79.34167
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® 8 Elite Gen 5	4096	11.220493	0.360336156 - 11.530757

Model	Runtime	Precision	Chipset	Context Length	Response Rate (tokens per second)	Time To First Token (range, seconds)
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X2 Elite	512	22.484254	0.12369325 - 0.12369325
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X2 Elite	4096	24.554353	0.447898468 - 14.332751
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X2 Elite	4096	24.309187	0.447912937 - 14.333214
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X2 Elite	4096	15.507219	0.193845843 - 6.203067
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X Elite	512	27.988389	0.40832025 - 0.40832025
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X Elite	512	27.143556	0.46132475 - 0.46132475
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X Elite	512	16.042448	0.25184725 - 0.25184725
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X Elite	4096	15.617096	0.913791375 - 29.241324
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X Elite	4096	16.000366	0.959076874 - 30.69045999

Model	Runtime	Precision	Chipset	Context Length	Response Rate (tokens per second)	Time To First Token (range, seconds)
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Snapdragon® X Elite	4096	9.673267	0.411740531 - 13.175697
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ IQ-X7181	512	27.988389	0.40832025 - 13.175697
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ IQ-X7181	512	27.143556	0.46132475 - 13.175697
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ IQ-X7181	512	16.042448	0.25184725 - 13.175697
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ Q-8750	512	24.055461	0.946880250 - 3.787521000
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ Q-8750	512	23.939526	0.95809125 - 3.787521000
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ Q-8750	512	19.557942	0.1830165 - 0.411740531
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ IQ-9075	4096	13.1	1.997191888 - 63.91014
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ IQ-9075	4096	3.0	2.471042471 - 79.073359

Model	Runtime	Precision	Chipset	Context Length	Response Rate (tokens per second)	Time To First Token (range, seconds)
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ IQ-9075	4096	10.0	0.441379309 - 14.124138
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ IQ-X7181	4096	15.617096	0.913791375 - 29.241324
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ IQ-X7181	4096	16.000366	0.959076874 - 30.69045999
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ IQ-X7181	4096	9.673267	0.411740531 - 13.175697
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ Q-8750	4096	13.965758	1.940274843 - 62.088795
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ Q-8750	4096	8.177678	2.355163312 - 75.3652259
Phi-4-Mini-Instruct	GENIEX_LLAMACPP	q4_0	Qualcomm® Dragonwing™ Q-8750	4096	13.116693	0.352867687 - 11.29176599

License

- The license for the original implementation of Phi-4-Mini-Instruct can be found [here](#).

References

- [Phi-4 Technical Report](#)
- [Source Model Implementation](#)

Community

- Join [our AI Hub Slack community](#) to collaborate, post questions and learn more about on-device AI.
- For questions or feedback please [reach out to us](#).

Usage and Limitations

This model may not be used for or in connection with any of the following applications:

- Accessing essential private and public services and benefits;
- Administration of justice and democratic processes;
- Assessing or recognizing the emotional state of a person;
- Biometric and biometrics-based systems, including categorization of persons based on sensitive characteristics;
- Education and vocational training;
- Employment and workers management;
- Exploitation of the vulnerabilities of persons resulting in harmful behavior;
- General purpose social scoring;
- Law enforcement;
- Management and operation of critical infrastructure;
- Migration, asylum and border control management;
- Predictive policing;
- Real-time remote biometric identification in public spaces;

- Recommender systems of social media platforms;
- Scraping of facial images (from the internet or otherwise); and/or
- Subliminal manipulation

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